

Brooke Proctor

blproctor@willamette.edu

(971) 720 8166

EDUCATION

Master of Data Science, Willamette University

August 2026

Bachelor of Science, Willamette University

May 2026

Major: Environmental Science and Data Science

SKILLS

Technical - R, Python, GIS, Pix4D, CloudCompare, Litchi, GlobalMapperPro, DJI for Drones

Libraries - Numpy, Scipy, Matplotlib, SciK

PROJECTS

Prey Population Predictions, Independent Study

August 2025 – December 2025

- Evaluated models with different tuning parameters to find the most efficient and accurate way to predict prey populations, leading to findings about extinction events and gaps in the prior research

Customer Churn Business Insights, Models and Machine Learning

January 2025 – May 2025

- Utilized and tuned five models to understand what factors encourage customers to leave a cell service provider, allowing for a marketing campaign to target specific at-risk customers to prevent churning

Sentiments of Generative AI, Survey Methodology and Design

January 2024 – May 2024

- Designed a survey to identify trends in the sentiments of generative AI at Willamette University, to better inform AI policies, leading to the adoption of Google's Gemini

EXPERIENCE

Geospatial Analyst, *Marion Polk Food Share*, Salem, Oregon

May 2026 – August 2026

- Collated data from several government agencies to create a series of maps about food insecurity across Marion-Polk Counties to aid the food share in the identification of food deserts and spatial patterns of food insecurity across the counties, resulting in the adoption of a new mobile pantry site to serve a previously under-resourced area

Aerial Researcher, *National Science Foundation*, Naxos, Greece

May 2024 – July 2024

- Documented historic and natural outdoor spaces through the LiDAR and photogrammetric modelling of ancient marble quarry extents to aid the Greek Ministry of Culture in classifying cultural hearths as protected lands to prevent new developments from destroying the quarries
- Created a self-owned research project by following a research question with a group of 4 peers to analyze ancient marble quarries to discover the methods and extents of quarrying done in ancient Greece to better understand technologies and cultural significances of the past

Lead Catalyst, *Office of Civic Engagement*, Salem, Oregon

August 2023 - May 2026

- Connected students with community organizations to do volunteer work and internships to support the development of Salem's community goals, such as housing for all, through facilitating student engagement and learning
- Produced dashboards twice a year to detail quantitative data about volunteer work to support reporting initiatives at the university, to receive further funding to spend in the community in the coming years, resulting in a generous private donation of approximately \$20,000

Data Science Tutor, *QUAD Center*, Salem, Oregon

August 2023 - May 2026

- Supported the professors in teaching elementary data science and computer science topics to students by hosting office hours and tutoring sessions to target marginalized students in computational fields and provide one-on-one support with difficult topics, as well as encouragement and advice for moving through the courses, resulting in recruitment of majors and more passing grades from students
- Attended weekly staff meetings to review statistics and figures about students' attendance at tutoring sessions and develop strategies to improve outreach to those struggling in a course, resulting in an overall increase in QUAD Center usage